Effective system administration

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Backups

- Backup is an additional copy of data that can be used for restore and recovery purposes
- This Backup copy can be created by:
  - Simply coping data (there can be one or more copies)
  - Mirroring data (the copy is always updated with whatever is written to the primary

Backup strategy

- A backup is put in place to avoid permanent data loss to ensure the integrity of stored data i.e. can get back to a previous version and build up the data correctly if current data found to be in error.
- A backup strategy needs a well documented and tested recovery procedure to ensure backup data is restored properly.
- Any backup strategy starts with a concept of data repository.
- The backup data needs to be stored and organized. It can be as simple as writing a list of all backups on a sheet of paper or generating a computerized catalog.

Backup purposes

- **Disaster Recovery-** Restores production data to an operational state after disaster
- **Operational -** Restore data in the event of data loss or logical corruptions that may occur during routine processing
- **Archival -** Preserve transaction records, email, and other business work products for regulatory compliance

Factors in recovery procedure

- A well documented procedure that allocates roles and responsibilities to staff.
- Availability of alternative hardware and accommodation if the originals have been destroyed.
- Availability of original software such as the customized operating system and application programs!
- That up-to-date backup data is readily available

TYPES OF BACKUPS

1. **Full backup**
   - As the name suggests, this refers to the process of copying everything that is considered important and that must not be lost.
   - This type of backup is the first copy and generally the most reliable copy, as it can normally be made without any need for additional tools.
2. **Incremental backup**
   - it involves making copies of the files by taking into account the changes made in them since the previous backup.
   - i.e if you add 2 new files after a full backup, incremental backup will only copy these.

Types of backups

3. **Differential backup**
   - A differential backup has the same basic structure as an incremental backup—in other words, it involves making copies only of new files or of files that underwent some kind of change.
   - However, with this backup model, all the files created since the original full backup will always be copied again.
   - The advantage that differential backup offers over incremental backups is a shorter restore time. When speed is important, such as in a disaster recovery scenario where downtime is unacceptable, rapid restores can be crucial.

Backup Technology options

- The types of media most commonly used for storing data have changed over the years.
- Backups have been variously done on punch card, floppy disk, optical media like CD, DVD and Blu-Ray, tape, external hard disk, cloud-based storage services, and more.
- Knowing how long you have to keep your backup copy will help you identify the right medium to use for the backup.

Backup Technology options

Medium	Date of invention	Lifespan	Capacity
HD	1956	5–10 years	From GB to TB
Floppy disk	1971	3–5 years	Hundreds of KB to a few MB
CD/CD-ROM	1979	25–50 years	80 minutes or 700 MB
MD (Mini Disc)	1991	25–50 years	60 minutes or 340 MB
DVD	1994/1995	25–50 years	4.7 GB
SD card	1994	10 years or more	A few MB to tens of GB
USB flash drive	2000	10 years or more	A few MB to tens of GB
SSD	1970–1990	10 years or more	From GB to TB

Physical Network Security

- Physical security requires that building site(s) be safeguarded in a way that minimizes the risk of resource theft and destruction
- Physical security controls are designed to prevent unauthorized personnel from gaining physical access to network components such as routers, cabling cupboards and so on.
- Controlled access, such as locks, biometric authentication and other devices, is essential in any organization.
- The physical plant must be satisfactorily secured to prevent those people who are not authorized to enter the site and use equipment from doing so.

Physical Threats (Examples)

- Examples of physical threats include:
  - Natural events (e.g., floods, earthquakes, and tornados)
  - Other environmental conditions (e.g., extreme temperatures, high humidity, heavy rains, and lightning)
  - Intentional acts of destruction (e.g., theft, vandalism, and arson)
  - Unintentionally destructive acts (e.g., spilled drinks, overloaded electrical outlets, and bad plumbing)

Layers of Physical Security

- Layers in Physical Security are implemented at the perimeter and are moving towards an asset.
- The layers are as follows:
  - Deterrence
  - Detection
  - Access control
  - Security

Layers of Physical Security

1. Deterrence

- The goal of Deterrence methods is to convince a potential attacker that a successful attack is not possible due to strong defenses.
- Deterrence methods are classified into 4 categories:
  - Physical Barriers: These include fences, walls, vehicle barriers, etc. They also act as a Psychological deterrent by defining the perimeter of the facility and making intrusion seem more difficult.
  - Combination Barriers: These are designed to defeat defined threats. This is a part of building codes as well as fire codes.

Layers of Physical Security

1. Deterrence
   - Natural Surveillance: In this architects seek to build places that are more open and visible to authorized users and security personnel so that attackers are unable to perform the unauthorized activity without being seen. For example- decreasing the amount of dense and tall vegetation.
   - Security Lighting: Doors, gates or other means of the entrance should be well lit as Intruders are less likely to enter well-lit areas. Keep mind to place lighting in a manner, that is difficult to tamper.

Layers of Physical Security

2. Detection

- If you are using the manual key control system, you have no way of knowing the exact timestamp of when an unauthorized user requested a key or has exceeded its time limit.
- Detection methods can of the following types:
  - Alarm Systems and Sensors: Alarm systems can be installed to alert security personnel in case of an attempt of unauthorized access. They consist of sensors like perimeter sensors, motion sensors, etc.
  - Video Surveillance: Surveillance cameras can be used for detection if an attack has already occurred and a camera is placed at the point of attack. Recorded video can be used

Layers of Physical Security

3. Access Control

- These methods are used to monitor and control the traffic through specific access points.
- Access Control includes the following methods:
  - Mechanical Access Control Systems: These includes gates, doors, locks, etc.
  - Electronic Access Control: These are used to monitor and control larger populations, controlling for user life cycles, dates and individual access points.
  - Identification System and access policies: These includes the use of policies, procedures and processes to manage the access into the restricted area.

Layers of Physical Security

4. Security Personnel

- They play a central role in all layers of security.
- They perform many functions like:
  - Administering electronic access control.
  - Responding to alarms.
  - Monitoring and analyzing video footage and many more

Countermeasures and Protection Techniques

1. Protection against Dumpster Diving

- Dumpster Diving is the process of finding some useful information about the person or business from the trash that can later be used for hacking purpose.
- Since the information is in the trash, it is not useful for the owner but deemed useful to the picker.
- To protect against it, you need to follow certain measures:
  - Ensure all important documents are shredded and they are still secure.
  - Destroy any CDs/ DVDs containing personal data.
  - Make sure that nobody can walk into your building and simply steal your garbage and should have safe disposal policy.
  - Firewalls can be used to prevent suspicious users from accessing the discarded data.

Countermeasures and Protection Techniques

2. Employee Awareness Training

- A negligent employee can be one of the major causes of a Cyber security breach. Employee awareness training sessions can help in such cases.
- Employees should be informed about the cyber risks that are there and how they can stay safe.

3. Site Access Control

- Lack of Access Control can be highly devastating if a wrong person gets in and gets access to sensitive information. Fortunately nowadays, you have a number of modern tools that will help you to optimize your access control.
- Envoy is a tool that will help you to expand access to guests in controlled manner.
- Open Path is a mobile system that allows access to only a limited set of people within the directory using smartphones and other devices.

Countermeasures and Protection Techniques

4. **Securing Your Windows**
   - If you have the data that hackers would love to get their hands on, they will try any method and might just look through the window.
   - Make sure you are aware of the sight angles to position your screens and other devices.

5. **Secure Network-Enabled Printers**

- Network Printers are a very convenient option allowing anyone in the office to get connected, without a need of extra wiring. Unfortunately, they have underlying security risks also.
- Sometimes, due to default settings, they offer open WiFi access, thus allowing anyone to get in and open vulnerabilities in the process.
  - Only connect those to the Internet that actually needs to be.
  - Remote access is not necessary for scenarios where only people from your office use the printer.
  - You can add passwords to the connection if necessary.

Countermeasures and Protection Techniques

6. **Securing Your Backups**

- Physical backups are critical for business continuity, helping you prevent data loss in the event of disasters, outages, and more.
- Most businesses secure their servers but they forget that backups are equally important. They are holding the same level of sensitive data as servers.
- Treat your backups as you treat your sensitive information and secure them.

8. Locking up your Servers

- Any area in your organization that stores data need to be secured. Locking doors and making sure server area gets extra protection.

Countermeasures and Protection Techniques

9. **Accounting for Loss or Stolen Devices**

- As devices are becoming more mobile, chances for them being stolen or falling out of someone's pocket becomes more frequent.
- Mobile Device Management can help you to manage such situations and take the necessary precautions.
- The best solution in such cases is to simply lock down and potentially wipe any lost or stolen devices from the organization remotely.

10. **Implementing video systems**

- To achieve a more secure premises, it is advisable to use a Video Surveillance system.
- Mere presence of cameras can deter potential attackers.
- Availability of video footage allows you to have continuous monitoring over the entire premises.
- If an attack happens, you can check the recorded video, easily reconcile the process and catch the perpetrator.